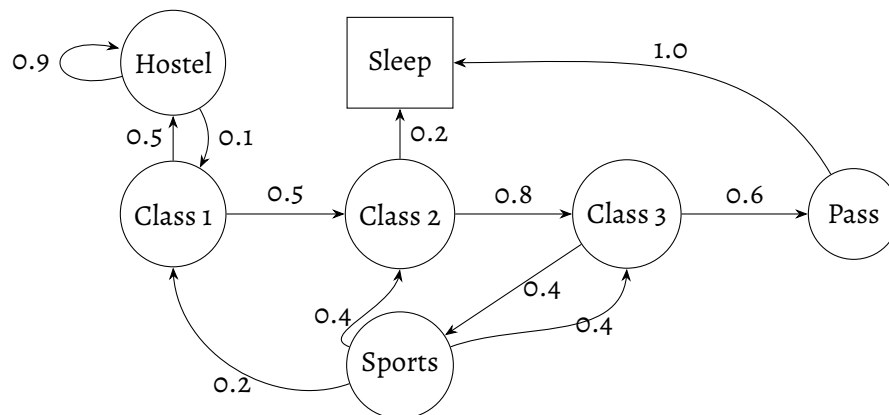


AI 3000: Reinforcement Learning

Tutorial 1

1 Understanding a Markov Chain Through an Example

Consider the following Markov chain representing a student's daily activities. Each node corresponds to a state, and every directed edge is labeled with the probability of moving to the next state in one time step. We use this example to understand the basic terminology and interpretation of Markov chains before studying their mathematical properties.



1. Answer the following questions using the Markov chain shown above.
 - (a) List the state space $\mathcal{S}$ of the Markov chain.
 - (b) Identify the absorbing state(s), if any. Justify your answer.
 - (c) Suppose the student is currently in *Class 2*. List all possible next states together with their one-step transition probabilities.
 - (d) Is it possible for the student to move from *Hostel* to *Class 2* in one step? Explain.
 - (e) Starting from the *Sports* state, list all possible one-step successor states.
 - (f) Construct the transition probability matrix of the Markov chain.
 - (g) Verify that your transition probability matrix is stochastic.
 - (h) Starting from *Hostel*, give one possible path that reaches the *Sleep* state.
 - (i) Explain why this process satisfies the Markov property. (Hint: Think of some trajectory and then show the Markov Property)
2. Let the initial distribution be $\mu = (1, 0, 0, 0, 0, 0, 0)$, corresponding to the student initially being in the *Hostel* state. Describe how you would generate a sample path

$$X_0, X_1, \dots, X_5$$

using the transition probability matrix of the Markov chain.

2 Markov Decision Processes

Consider the following time-homogeneous MDP.

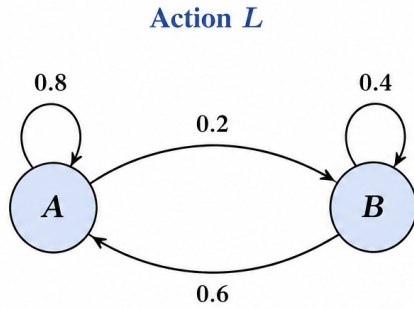


Figure 1: State transition graph when action L is taken.

The numbers on the arrows denote the transition probabilities $P(s' | s, L)$. For example, $P(A | A, L) = 0.8$, $P(B | A, L) = 0.2$, $P(A | B, L) = 0.6$, and $P(B | B, L) = 0.4$.

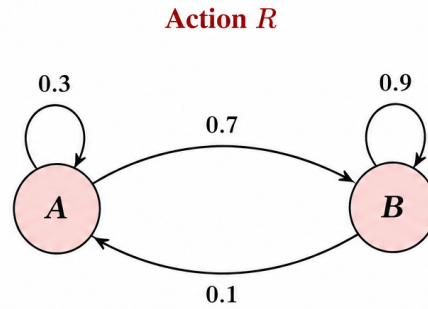


Figure 2: State transition graph when action R is taken.

The numbers on the arrows denote the transition probabilities $P(s' | s, R)$. For example, $P(A | A, R) = 0.3$, $P(B | A, R) = 0.7$, $P(A | B, R) = 0.1$, and $P(B | B, R) = 0.9$.

Note: The environment (transition probabilities) is the same at all time steps (time-homogeneous). The two graphs above show the transition dynamics under each action separately.

1. Assume the stationary randomized Markov policy

$$\pi(L|A) = 0.7, \quad \pi(R|A) = 0.3,$$

$$\pi(L|B) = 0.4, \quad \pi(R|B) = 0.6.$$

- (a) Compute $P^\pi(S_{n+1} = B | S_n = A)$.
- (b) Compute $P^\pi(S_{n+1} = A | S_n = B)$.
- (c) Construct the induced state transition matrix.
- (d) First compute $P^\pi(B, R | A, L)$. Construct the transition probability matrix of the state-action process $\{(S_n, A_n)\}$.
- (e) Is the process $\{(S_n, A_n)\}$ time homogeneous? Justify your answer.

2. Now suppose the policy is

$$\pi_{2k}(L|s) = 1, \quad \pi_{2k+1}(R|s) = 1,$$

for every state s . Answer the following.

- (a) Show that $\{(S_n, A_n)\}$ is a Markov chain.
- (b) Compute the one-step transition probabilities from (A, L) at times $n = 0$ and $n = 1$.
- (c) Is the state-action process homogeneous or inhomogeneous? Explain.

3 History-Dependent Policy: A Counterexample

Consider the same time-homogeneous MDP used in the previous questions.

Suppose the agent now follows the following **history-dependent deterministic policy**:

$$A_n = \begin{cases} L, & \text{if } S_{n-1} = A, \\ R, & \text{if } S_{n-1} = B, \end{cases} \quad n \geq 1.$$

That is, the action at time n depends on the *previous state* rather than only on the current state.

Answer the following questions.

1. Compute

$$P(S_{n+1} = B \mid S_n = A, S_{n-1} = A).$$

2. Compute

$$P(S_{n+1} = B \mid S_n = A, S_{n-1} = B).$$

3. Compare the two probabilities obtained above.

4. Is the state process

$$\{S_n\}_{n \geq 0}$$

a Markov chain? Justify your answer.

5. Is the state process

$$\{S_n, A_n\}_{n \geq 0}$$

a Markov chain? Justify your answer.

6. Which assumption of Theorem 1 is violated in this example? Explain why this assumption is essential.

Hint: Recall that under a Markov chain,

$$P(S_{n+1} \mid S_n, S_{n-1}, \dots, S_0) = P(S_{n+1} \mid S_n).$$